Exercise Sheet 1

[DATA23UF] Probability and Statistics for Data Science

October 15, 2026

1. 🌶🌶🌶 Two fair dice are rolled. Consider the Laplace experiment

$$\Omega = \{1, \dots, 6\}^2, \qquad \mathcal{A} = 2^{\Omega}, \qquad \mathbb{P}[\{\omega\}] = \frac{1}{36} \quad (\omega \in \Omega),$$

and define the random variable $X(i, j) := i + j$.

a) Determine whether each of the following events belongs to $\sigma(X)$:

$$A := \{(i, j) \in \Omega : i + j \geq 8\},$$
$$B := \{(i, j) \in \Omega : i > j\},$$
$$C := \{(i, j) \in \Omega : i + j \text{ is even}\}.$$

Justify your answers.

b) Define

$$Y := \mathbb{1}_{\{X \text{ is even}\}}.$$

Describe $\sigma(Y)$ and verify that

$$\sigma(Y) \subseteq \sigma(X).$$

Is the inclusion strict?

2. 🌶🌶🌶 Let X be an $\mathbb{N}$-valued random variable with probability mass function

$$p_X(k) = \frac{c}{k(k+1)}, \qquad k \in \mathbb{N},$$

where $c > 0$ is unknown.

a) Determine c.

b) Determine $\mathbb{P}[X \geq m]$ for every $m \in \mathbb{N}$.

c) Sketch the cumulative distribution function F_X.

d) Compute $\mathbb{P}[3 < X \leq 8]$.

3. 🌶🌶🌶 Let X be a real-valued random variable and define

$$F_X(x) := \mathbb{P}[X \leq x], \qquad x \in \mathbb{R}.$$

Using only the basic properties of probability measures from Proposition 1.3, prove the following statements formally:

a) F_X is non-decreasing.

b) F_X is right-continuous.

c) $\lim_{x\to\infty} F_X(x) = 1$.

d) $\lim_{x\to-\infty} F_X(x) = 0$.

e) For every $x \in \mathbb{R}$,

$$\mathbb{P}[X = x] = F_X(x) - F_X(x^-),$$

where $F_X(x^-) := \lim_{y\uparrow x} F_X(y)$.

4. 🌶🌶🌶 Consider the function $F : \mathbb{R} \to [0,1]$ given by

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{3} + \frac{2x}{3}, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

a) Verify that F satisfies the properties of a cumulative distribution function.

b) Let X be a random variable with CDF F. Determine $\mathbb{P}[X = 0]$.

c) Compute $\mathbb{P}[\frac{1}{4} < X \leq \frac{3}{4}]$.

d) Can the law of X be represented completely by a probability mass function? Explain.

e) Can there exist a probability density function f_X such that

$$\mathbb{P}[X \in B] = \int_B f_X(x)\,\mathrm{d}x$$

for every Borel set $B \subseteq \mathbb{R}$? Explain.